

ACPollerForMail

2.0

Manual

Mail capture, PDF conversion
and delivery to ECM

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ACPollerForMail — Manual

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1. What the product does

ACPollerForMail watches mailboxes, converts everything that arrives into PDF, and delivers the result to an ECM system or a file share.

The cycle, for each message:

1. **Collection** over IMAP, Microsoft Graph, or by reading a folder
2. **Parsing** of the message into a tree: body, attachments, archives, embedded messages
3. **Conversion** of every part into PDF
4. **Assembly** into a single document, with one bookmark per part
5. **Information file** describing the message and its contents
6. **Delivery** to one or more targets
7. **Marking** the message as processed

The order of those last two steps is not negotiable: marking only happens once delivery is confirmed. A duplicate can be spotted and corrected, a silent loss cannot.

What it does not do

It does not read document contents: no character recognition, no data extraction. Those belong to business plugins, covered in chapter 9.

It does not send mail, except acknowledgements when a plugin asks for them.

2. Concepts worth knowing

Configuration

A **configuration** describes one mailbox to watch and what to do with its contents: where to collect from, how to convert, where to deliver. Each configuration runs its own *worker*, independently of the others.

A faulty configuration is **set aside**: the others keep running. This is deliberate, and it is what allows a fleet of a hundred mailboxes to be operated without one typing mistake bringing everything down.

Template

A **template** is a model configuration that others inherit from. Across a fleet sharing the same Microsoft 365 tenant, the same credentials and the same targets, each mailbox comes down to three lines: its name, the template, its address.

The day the client secret changes, one edit is enough.

A configuration may override any inherited value. One rule is worth remembering: **lists are replaced, never merged**. Redeclaring targets in a configuration replaces the template's entirely.

Idempotence

The service keeps a log of messages already processed, in a local database. A message already delivered is never processed again, even after a restart, even if the message is still sitting in the mailbox.

This log is **the most critical file in the product**. Losing it causes everything still in the mailboxes to be processed again, hence duplicates in the ECM.

Working unit

Each message being processed occupies a **working** directory holding the original message, the extracted parts and the produced PDFs. It is only cleaned up once delivery is confirmed.

A build-up in that directory therefore points to failed deliveries, not to a cleanup defect.

Part status

The information file carries a status per part. Telling them apart matters: three of them mean the ECM does not have the document.

Status	Meaning
Converted	Faithful PDF produced, present in the merged document
PassThrough	Already a PDF, taken as is or repaired
PassThroughUnverified	Delivered alongside, ABSENT from the merged document
Substituted	Not converted : a page carrying the reason replaces it
Failed	Nothing reaches the ECM for this part
Excluded	Deliberately dropped: filter or size threshold
Container	Archive or embedded message: only its children count

3. Architecture

The product is made of two distinct applications.

The Windows service does all the work. It runs without an open session, starts automatically, and restarts by itself after a failure.

The monitoring interface does nothing by itself: it talks to the service over a local API. It can be closed without consequence, and installed on a separate workstation.

That separation has a practical consequence: the interface never writes directly into the configuration files. The service stays the sole owner of what it reads and when it re-reads it. An interface writing the file while a worker reads it would produce an inconsistent state, with no error.

External components

Component	Role	Without it
LibreOffice	Converting office documents	Word, Excel and PowerPoint come out as substitution pages
qpdf	Repairing unreadable PDFs	A few percent of invoices are not merged

Both are called out of process, never linked into the product. Their absence degrades the service, it does not stop it. —

4. Installation

Prerequisites

Item	Requirement
Operating system	Windows Server 2016 or later
.NET runtime	None: the build is self-contained
Disk space	500 MB, plus the working volume
Service account	Dedicated domain account, see below

Windows Server 2012 R2 is not supported.

The service account

This is the installation decision with the heaviest consequences.

The Local System account has NO network identity. A service running under it can neither reach a UNC share nor authenticate against an internal FTP server. The error only surfaces at the first delivery, hours after installation.

The dedicated domain account needs:

- **Log on as a service**, in the local policy
- **Modify** on the ECM delivery folder
- **Read** on the certificate private key, if Graph authentication uses a certificate

It does not need to be a server administrator.

Running the installer

1. Run `ACPollerForMail-x.y.z-setup.exe` as administrator
2. Pick the components: LibreOffice and qpdf if the server lacks them
3. **Enter the service account** on the dedicated page
4. Let the wizard finish

The installer creates the directories, generates the control token, sets the NTFS permissions, and registers the service as delayed automatic start with automatic restart after failure.

Installing LibreOffice silently takes several minutes. The progress page looks frozen during that time.

After the first start

Secrets entered in clear text are encrypted on first access, and the service keeps a copy of the original file.

Those copies hold the secrets IN CLEAR TEXT. Check that encryption took place, then delete them:

```
findstr /i "ENC:" "C:\Program Files\ACPollerForMail\appsettings.json"
del "C:\Program Files\ACPollerForMail\appsettings.json.clear.*.bak"
```

The service purges them by itself after twenty-four hours, but there is no reason to wait.

5. The interface

The status bar

The first thing to look at. A green light and the number of active workers mean the service is running. A red light tells two cases apart by its message: service unreachable, or token rejected.

The two lists on the left

Configurations lists the watched mailboxes, with their protocol and the number of issues found. A blocking issue sets the configuration aside: it shows in red and its worker does not run.

Templates lists the models, with the number of configurations inheriting from each. That number is the **reach** of a change: seeing it before opening a template avoids editing fifty mailboxes while believing you are editing one.

The two tabs on the right

Monitoring shows the details of the selected configuration: its issues, the result of the last test, the state of every worker.

Dashboard shows activity over the last twenty-four hours.

The dashboard

Four indicators are enough to know whether things are fine.

Indicator	What it answers
Success rate	Is it running normally?
95th percentile duration	Are slow messages saturating the workers?
Graph throttling	Am I close to the tenant quota?
Free disk	How long before it stops?

The per-configuration breakdown at the bottom sorts mailboxes by failure count: the first row names the one causing trouble.

The 95th percentile deserves a word. On processing durations, the average hides the slow cases, and those are the ones holding concurrent slots while other mailboxes wait. A wide gap between the average and that percentile points to a handful of messages monopolising the service.

The buttons

Button	Effect
New, Edit, Delete	Act on the selected configuration
Test	Checks the source and every target, without processing a message
Restart	Restarts that configuration's worker, leaving the others alone
Refresh	Forces an update, also available on F5
Maintenance	Triggers an immediate cleanup pass
Service	Server settings, which require a restart
Connection	Address and token of the interface itself
Help	Token, status and troubleshooting reference, also on F1

Test is the button to use before any save. It actually writes to file targets, checking permissions and not merely the existence of the path. —

6. Configuring a mailbox

Each tab of the editor covers one stage of processing.

General tab

Name: unique identifier, used as the key in logs and as the working directory name. Changing it amounts to creating a new configuration, and the processed-message log starts from scratch.

Template: the model to inherit from. Empty for a standalone configuration.

Interval between cycles: delay between the end of one cycle and the start of the next. Too short a value across a large fleet multiplies calls without gaining anything, collection already being bounded by global concurrency.

Random startup delay: without it, every mailbox hits the server on the same second when the service starts, triggering throttling before the first message.

First-cycle lookback: applies to the very first cycle only, when nothing has been processed yet for this mailbox. Too low a value makes it look like nothing is arriving while the service works fine, and nothing reports it.

Floor date: no older message will ever be processed. Unlike the lookback, it does not slide. **Set it at go-live** when history is not being imported: without it, the first run processes everything sitting in the mailbox.

Protocol tab

Choosing the protocol reveals the matching fields.

graph for Microsoft 365. **Certificate** authentication is preferable to a client secret, which always expires on a Friday evening.

imap for a classic mail server. Port 993 means implicit SSL, port 143 means STARTTLS.

folder to read `.eml` files from disk. This mode serves to replay a problematic message and to run acceptance tests, without a mailbox or a network.

Watched folder: designated by its canonical name, not its label. On a French mailbox the label is *Boîte de réception*, and any lookup by label would fail.

After processing: applied once delivery is confirmed. `None` leaves the message untouched, which allows unlimited replays during acceptance testing. In production, `MarkAsRead` or `Move` avoid listing the same messages forever.

Throttling, Graph only. These bounds are **shared by every mailbox of the same tenant**. Spacing matters as much as concurrency: without it, eight calls leave on the same millisecond and trigger throttling before reaching the limit.

Conversion tab

Timeout per part: once exceeded, the converter is **killed** rather than waited for. A LibreOffice instance stuck on a corrupt document never recovers.

Unconvertible part: three policies.

Policy	Effect
<code>Substitute</code>	A page carrying the name and the reason replaces the part
<code>Skip</code>	The part is dropped, leaving no trace in the PDF
<code>RejectMessage</code>	The whole message is rejected, no partial output

`Substitute` is the recommended default: the ECM sees something is missing, which triggers a supplier follow-up. With `Skip`, the gap goes unnoticed.

Inline image threshold: below it, an image referenced by the HTML body is dropped. Without this filter, every message produces three pages of signature logos.

Extraction bounds: depth, part count, expanded volume, single entry size. They protect against a crafted archive, and processed files arrive by mail, therefore from anyone. Raising them without reason exposes the service.

Excluded extensions: used to drop electronic signatures and calendar invitations, which have no business reaching the ECM.

Output format tab

PDF production mode:

Mode	Result
Merged	One PDF per message, body then parts, with bookmarks
PerAttachment	One PDF per part
Both	Both of the above

Information file format: `json`, `xml`, `csv` or `xslt`.

`xslt` applies a transformation sheet to the canonical XML. This is what makes it possible to produce the exact structure an ECM expects, including flat text, without shipping code. To develop a sheet: set the format to `xml`, look at the actual output, write the sheet against it, then switch back to `xslt`.

Naming template: see appendix A for the token list. In `PerAttachment` mode the `{index}` token is **mandatory**, otherwise parts overwrite each other.

Atomic write: the file is written under a temporary extension, then renamed. Without it, an ECM watching the folder can pick up a PDF mid-write, and the error is undetectable on its side.

All or nothing across targets: if one target fails, deliveries already made to the others are rolled back. An ECM holding a message that another one never received creates a gap nobody notices before the audit.

Sentinel file: written last, it tells the ECM the batch is complete. Without it, the ECM may take the PDF without its information file, or the other way round.

Processors tab

The list of business processors provided by the installed plugins, with the pipeline stages at which they run.

A processor framed in red is declared in the configuration but its plugin is no longer loaded: it will not run, and the service will set the configuration aside at the next start.

Rejected plugins appear below with their reason. Without that list, one would hunt for a processor that does exist in the folder but whose plugin was never loaded.

Targets tab

A configuration may deliver to several targets. Each target carries a **name** that appears in the logs: without distinct names, a failure on one of three targets is unreadable.

The fields offered depend on the transport, but the table at the bottom shows **every** setting, including those no field covers. That is what makes it possible to configure a customer export DLL.

Fields tab

Fields added to the information file. **Order matters:** it is the CSV column order, and changing it can break an integration already in place.

See appendix B for the available sources and the resolution order.

7. Delivery targets

File system

The most common transport: a local folder or a UNC share.

The service account needs **modify** permission on that folder. A UNC share further requires the service to run under a domain account, the Local System account having no network identity.

The **Test** button actually writes a witness file, then deletes it: it checks permissions, not merely the existence of the path.

FTP and FTPS

Explicit encryption on port 21 is the default, implicit uses port 990.

Accept any certificate removes every protection against interception: a third party can sit between the service and the server undetected. Enable only against a known self-signed certificate, and document it. The service repeats the warning in its log on every session.

S3 and compatible stores

For AWS, the region is enough. For MinIO, Garage or Ceph, fill in the **custom endpoint**: path style and the compatible checksum mode are then enabled automatically. Without them, those implementations reject deliveries with an error that does not name the cause.

Server-side encryption must stay empty if the store does not configure it: declaring it would make every delivery fail.

Keys may be left empty to use the usual AWS resolution chain, environment variables or instance role.

8. Custom fields and transformation

Why

No fixed format suits every ECM: one wants a hard-coded company code, another the current date in its own format, a third a header set by its relay. Freezing these in code would mean one release per customer.

The five sources

Source	Value
Fixed	Literal, identical for every message
Token	Token template, resolved like file naming
Header	MIME header, name case-insensitive
Property	Value left behind by a business processor
Environment	Server environment variable

Resolution order

Source value, then date format, then translation table, then default value if the result is empty, then fixed length.

The **translation table** turns a label into the code the ECM expects. A value missing from the table passes through unchanged.

Fixed length serves ECMs reading positional formats. A value that is too long is truncated: overflowing would shift every following column, which is worse than losing the end of a label.

An unresolved field **never fails the processing**. It produces its default value and a warning. The document is what matters, not the completeness of its index.

XSLT transformation

The service always produces the same canonical XML. A customer-specific style sheet turns it into whatever their ECM expects.

The output need not be XML: a sheet declaring `xsl:output method="text"` produces flat, CSV or fixed-width output.

For safety, **scripts and the `document()` function are disabled**. A sheet is a configuration file, and allowing code execution from a configuration file would amount to granting the service account's rights to anyone able to write in that folder.

9. Business processors

A plugin is a DLL dropped into the plugin folder. The service reads it by metadata without executing its code, checks compatibility, then loads it into an isolated context.

A plugin may provide:

- **processors** running at various pipeline stages
- **delivery targets** specific to the customer
- **reference data providers** consulted by processors

Stages

Stage	When
AfterParse	After the message is parsed, before conversion
BeforeConvert	Before each part conversion
AfterConvert	After conversion, before assembly
BeforeExport	Before delivery, last chance to reject
AfterExport	After delivery is confirmed

A processor may **reject** a message or **skip** it. Rejection reports an anomaly, skipping marks a message that does not concern this flow. Both appear separately on the dashboard.

A processor leaves its results in a property bag, which custom fields of source `Property` can pick up. That is how a supplier code extracted from the subject line reaches the information file.

Contract version

Every plugin declares the contract version it expects. A plugin that is too old is **rejected and not loaded**, with its reason visible in the interface. Loading it would produce runtime errors, far harder to diagnose than an explicit refusal at start-up.

10. Service settings

The **Service** button. These settings are **not reloadable while running**: any change requires a full restart, during which capture is interrupted.

After saving, an amber banner appears and a notification shows in the system tray. The details list the pending settings and the reason for each.

What they cover

Tab	Settings
Service	Working directory, global concurrency, plugins, retention
Tools	LibreOffice, profiles, qpdf, fonts
Maintenance	Interval, quarantine, disk threshold
Control	API enablement and port, token, metrics database

Three traps

Changing the working directory loses the idempotence log. Messages still in the mailboxes will be processed again.

Raising global concurrency beyond the LibreOffice profile count achieves nothing: conversion becomes the bottleneck.

Changing the token invalidates every operator workstation, including the one it was changed from.

Restarting

The interface does not restart the service itself: it talks to it over its API, and a service shutting down on its own client's order could not confirm the order was carried out.

Restart-Service ACPollerForMail

11. Day-to-day operation

Checking that things are fine

Open the interface, Dashboard tab. If the success rate is 100 %, no Graph throttling shows and the disk has room, there is nothing to do.

Without the interface:

```
curl -H "X-ACPollerForMail-Token: <token>" http://127.0.0.1:5199/api/status
```

Suspending a mailbox

Clear **Configuration active**, save. The worker stops, the settings are kept.

Changing a password

Protocol tab, enter the new value, save. The secret is encrypted by the service on first access.

Leaving the field empty keeps the current value. The interface never receives secrets, so it cannot send them back.

Adding a mailbox to an existing fleet

Use a template. The new configuration comes down to its name, the template and the address.

Replaying a message

The original message is kept in its working directory under the name `message.eml`, and in quarantine for abandoned units.

1. Copy the `.eml` into a folder
2. Create a configuration using the **folder** protocol pointing at it
3. Give it test targets
4. Start it

The message is replayed identically, without touching the mailbox or the network.

Note: the processed-message log applies in folder mode too. To replay the same message twice, change the configuration name.

Resuming after a long outage

After several days down, the mailboxes hold the backlog.

1. Check disk space before starting
2. Temporarily lower the per-cycle message cap
3. Watch Graph throttling during the first hour
4. Restore nominal values once the backlog is cleared

12. Troubleshooting

A message comes back every cycle

First check that **delivery succeeds**: marking only happens once delivery is confirmed.

If delivery succeeds, look at the disposition: with `None` the message stays unread and will be listed again every cycle.

The merged PDF is empty or incomplete

Open the information file and look at the status of each part. Three statuses mean the ECM does not have the document: `PassThroughUnverified`, `Substituted` and `Failed`.

If every office document fails, LibreOffice is missing or misconfigured.

Nothing is collected, with no error

The service runs and finds nothing. Look, in this order:

1. **First-cycle lookback**: a low value excludes older messages
2. **Unread only**: messages opened in the webmail are marked as read
3. **Floor date**: later than the messages present
4. **Watched folder**: messages may sit in a subfolder

Switching the log level to `Debug` makes the cycle verbose:

```
setx ACPollerForMail_LOGLEVEL Debug /M
Restart-Service ACPollerForMail
```

The disk fills up

The working directory is only cleaned up once delivery is confirmed. A build-up points to failed deliveries.

The **Maintenance** button triggers an immediate pass: quarantine of unfinished units older than seven days, removal of orphan directories.

The service does not start

```
findstr /i "worker(s) démarre" logs\ACPollerForMail-*.log
```

The line `N worker(s) démarre(s)`, `M ecarte(s)` gives the count. If `M` is not zero, the reasons are just above, at `Error` level.

The interface says the token is rejected

The interface token does not match the service one. Open **Connection**, then **Test connection**: the message tells an unreachable service apart from a rejected token.

The token in clear text sits in `C:\ProgramData\ACPollerForMail\ui-settings.default.json`.

Tracing one message

The information file carries a `CorrelationId`, present in every log line of that processing:

```
findstr /i "<CorrelationId>" logs\ACPollerForMail-*.log
```

13. What not to do

Do not delete the processed-message log to start clean. Everything still in the mailboxes will be processed again.

Do not edit the configuration by hand while the service runs. The interface detects concurrent changes and refuses to overwrite, but the reverse is not true.

Do not delete `.bak` backups without reading them. They are the automatic copies taken before each change, and the only way back. The `.clear.*.bak` files, on the other hand, hold secrets in clear text and must be deleted once encryption is verified.

Do not raise the extraction bounds without reason. They protect against a crafted archive, and processed files arrive from anyone.

Do not forget the floor date at go-live. Without it, the first run processes all the history sitting in the mailbox. —

Appendix A — Naming tokens

Usable in the naming template and in fields of source `Token` . A format may follow a colon, for example `{received:yyyyMMdd}` .

Token	Value
<code>{date}</code>	Processing date, in local time
<code>{time}</code>	Processing time
<code>{datetime}</code>	Processing date and time
<code>{received}</code>	Message reception date
<code>{guid}</code>	Random identifier, guarantees uniqueness
<code>{mailbox}</code>	Local part of the mailbox address
<code>{config}</code>	Configuration name
<code>{subject}</code>	Message subject, truncated and sanitised
<code>{from}</code>	Local part of the sender address
<code>{index}</code>	Part number, mandatory in PerAttachment mode
<code>{name}</code>	File name without extension
<code>{ext}</code>	Part extension
<code>{node}</code>	Full logical path

`{date}` changes if the message is replayed the next day, `{received}` does not. For naming that must stay reproducible after an incident, prefer `{received}` .

Every character forbidden in a Windows file name is replaced, path separators included: a subject containing a slash cannot create a subfolder on the target.

Appendix B — Custom field output

The same content comes out differently depending on the format.

json: a `fields` object, names as keys.

xml: a `Fields` block, the field name being an **attribute** rather than an element name. A name typed by an operator may contain a space or start with a digit, which would produce invalid XML as an element name.

csv: columns appended **after** the fixed ones, repeated on every row. Adding a field therefore never shifts the existing columns.

xslt: available to the transformation sheet.

Appendix C — Locations

Path	Contents	Critical
C:\Program Files\ACPollerForMail\appsettings.json	Configuration, encrypted secrets	Yes
C:\Program Files\ACPollerForMail\logs\	Logs	No
C:\Program Files\ACPollerForMail\docs\	This manual	No
C:\ProgramData\ACPollerForMail\work\	Working units in progress	No
C:\ProgramData\ACPollerForMail\work\state\processed.db	Processed-message log	Critical
C:\ProgramData\ACPollerForMail\work_quarantaine\	Abandoned units, 30 days	No
C:\ProgramData\ACPollerForMail\ui-settings.default.json	Interface address and token	No
%APPDATA%\ACPollerForMail\ui-settings.json	Operator's own settings	No

Backup

Two items belong in the server backup:

- C:\ProgramData\ACPollerForMail\work\state\processed.db
- C:\Program Files\ACPollerForMail\appsettings.json

The first is the only one whose loss is irreversible. The second can be rebuilt, but every secret would have to be entered again.

Appendix D — Control API

Listens on the loopback interface only. Every request carries the `X-ACPollerForMail-Token` header.

Method	Route	Effect
GET	<code>/api/status</code>	Service and worker state
GET	<code>/api/dashboard</code>	Indicators over the last 24 hours
GET	<code>/api/configurations</code>	Configurations and their issues
GET	<code>/api/configurations/{name}/raw</code>	One configuration as JSON, secrets masked
PUT	<code>/api/configurations/{name}</code>	Creates or replaces a configuration
DELETE	<code>/api/configurations/{name}</code>	Deletes a configuration
POST	<code>/api/configurations/{name}/test</code>	Tests source and targets
POST	<code>/api/workers/{name}/restart</code>	Restarts one worker
GET	<code>/api/templates</code>	Templates and their reach
GET	<code>/api/processors</code>	Available processors, rejected plugins
GET	<code>/api/settings</code>	Service settings, secrets masked
PUT	<code>/api/settings</code>	Changes service settings
POST	<code>/api/maintenance/run</code>	Triggers a maintenance pass

Secrets **never** leave the API: they are replaced by an empty string. On write, an empty field means unchanged.

Every read returns a **fingerprint** of the configuration file, which writes demand back. If the file changed in between, the change is refused rather than overwriting someone else's.

Appendix E — Go-live checklist

To be run on the customer server before declaring the service operational.

Check	How	Expected
Service started	<code>Get-Service ACPollerForMail</code>	Running
Active workers	Start-up log	N started, 0 set aside
Source reachable	Test button	Every component green
Targets reachable	Test button	Actual write confirmed
LibreOffice	Start-up log	Path resolved
qpdf	Start-up log	Path resolved
Account network access	Test a UNC target	Delivery succeeds
First message	Drop a test message	PDF and information file delivered
Idempotence	Wait for a second cycle	The message is not processed again
Restart	<code>Restart-Service ACPollerForMail</code>	Resumes without duplicates
Floor date	General tab	Set if history is not imported
Encrypted secrets	<code>findstr ENC: appsettings.json</code>	At least one match
Clear-text backups	<code>dir *.clear.*.bak</code>	Deleted

The **account network access** row is the one that fails most often, and it only fails against a real share: a local test does not cover it.